

Notes: Inderst and Muller (JFE, 2004)

The Effect of Capital Market Characteristics on the Value of Start-Up Firms

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 This is a theory paper: no estimation sample
- 2.2 Core economic environment
- 2.3 Project cash flows and effort
- 2.4 Ownership and utility objects
- 2.5 Efficient sharing rule
- 2.6 Search-market objects
- 2.7 Outside options and delay
- 2.8 Entry cost and transparency
- 2.9 Valuation objects
- 2.10 Screening extension

3 Models and empirical specifications (all equations + interpretation)

- 3.1 Financial contracting and the double-sided incentive problem
- 3.2 From the equity frontier to the contract frontier
- 3.3 Generalized Nash bargaining
- 3.4 Search market and endogenous outside options
- 3.5 Capital-market equilibrium
- 3.6 Baseline comparative statics with respect to capital-market competition
- 3.7 Valuation formulas and why valuation is not the same as value
- 3.8 Market value and success probability
- 3.9 Endogenous entry of venture capitalists

3.10	Changes in investment profitability	
3.11	Changes in entry costs	
3.12	Changes in transparency	
3.13	Screening extension	
4	Identification and credibility (what makes the model persuasive)	
4.1	Why the paper departs from the “competitive VC market” benchmark	
4.2	Why ownership shares matter for real efficiency	
4.3	Why the model is internally disciplined	
4.4	Why free entry does not fix the problem	
4.5	Robustness to richer contracts	
4.6	Robustness to broader market structure	
4.7	Where the paper is strongest empirically	
4.8	Main limitations	
4.9	Relation to the literature	
5	Tables and figures	
5.1	Figure 1: equity frontier and contract frontier	
5.2	Proposition 1 and Figure 2: value creation versus valuation	
5.3	Proposition 2: pre- and post-money valuations	
5.4	Proposition 3: success probability and market value	
5.5	Figure 3 and Propositions 5–7: boom-bust dynamics	
5.6	Figures 4 and 5: safe payoff and wage extensions	
5.7	Section 7 empirical implications: what should move in the data	
6	Short summary	
7	Conclusion	
7.1	Core conclusion (what the paper establishes)	
7.2	Economic intuition	
7.3	Takeaway	

1 Big picture

- **Question.** How do venture-capital market conditions – capital supply, competition among investors, expected investment profitability, entry costs, and market transparency – affect start-up ownership shares, valuations, incentives, and the value actually created inside new firms?
- **Setting.** The paper builds an equilibrium model of venture-capital finance with *double-sided moral hazard*: both the entrepreneur and the venture capitalist must exert non-contractible effort for the project to succeed.
- **Core challenge.** Standard contracting models often assume a perfectly competitive capital market, so entrepreneurs effectively get all the surplus. That misses a central real-world fact: bargaining power shifts back and forth depending on whether capital or good projects are relatively scarce.
- **Main idea.** Capital-market conditions determine outside options in the search market. Outside options determine bargaining power. Bargaining power determines the venture capitalist's equity share. That equity share then shapes incentives, screening effort, and total value creation.
- **Key mechanism.** If venture capital is abundant, entrepreneurs can more easily walk away and search again. Their outside option improves, the venture capitalist's share falls, and founders keep more equity. That raises entrepreneurial incentives but weakens venture-capitalist incentives. Because both sides matter, value creation is generally *hump-shaped* in market competition.
- **Main theoretical contribution.** The paper ties three objects together in one equilibrium system: *pricing* (valuations and ownership shares), *contracting* (incentive balance between entrepreneur and investor), and *search* (how fast each side finds the other).
- **Main baseline result.** More competition among investors lowers the venture capitalist's ownership share and raises pre- and post-money valuations, but total value creation first rises and then falls. Too little competition gives investors too much equity; too much competition gives them too little.
- **Industry-dynamics result.** A rise in expected investment profitability leads to entry by venture capitalists in the long run. That increases capital-market competition, raises valuations, lowers investor ownership shares, speeds financing for entrepreneurs, and slows deal finding for investors.
- **Transparency result.** Better market transparency improves matching efficiency. In the long run it shifts bargaining power toward entrepreneurs and raises the value created in start-ups by reducing delay and intensifying competition.
- **Screening result.** Venture capitalists screen more in “down markets” when competition is weak, because walking away from a bad project is especially valuable when the investor's outside option is strong.
- **Why the paper mattered.** It offered a clean theoretical interpretation of the dot-com boom and bust: rising perceived profitability triggers entry, valuations rise, founder terms improve, and incentives can eventually become distorted when too much money chases deals.

- **Takeaway.** The paper’s core message is that the supply of venture capital changes not only prices but also *real efficiency*. Market structure affects ownership, ownership affects incentives, and incentives affect the success and value of start-ups.

2 Data and measurement (key objects)

2.1 This is a theory paper: no estimation sample

There is no micro dataset or regression design in the main analysis. Instead, the paper builds a stylized equilibrium model and studies comparative statics. The “measurement” objects are therefore the model’s primitives, state variables, and endogenous outcomes.

2.2 Core economic environment

- A **penniless entrepreneur** has a project that requires outside financing.
- A **venture capitalist** can finance the project and also add value through advice, monitoring, or support.
- Both sides must exert effort, so the contracting problem is not one-sided.

Interpretation. The investor is not just a passive financier. Venture capital ownership matters because it affects the investor’s own incentive to help the firm succeed.

2.3 Project cash flows and effort

The baseline project requires investment $I > 0$. Payoff is

- $X_h > I$ with success probability $p(e, a)$,
- $X_l = 0$ in the baseline failure state.

Here:

- $e \in [0, 1]$ is the entrepreneur’s non-contractible effort,
- $a \in [0, 1]$ is the venture capitalist’s non-contractible effort,
- $c(e)$ and $g(a)$ are strictly convex effort-cost functions.

Interpretation. Because success depends on both parties, the ownership split is fundamentally about balancing *two* incentive problems, not just paying back capital.

2.4 Ownership and utility objects

The venture capitalist’s ownership share is $s \in [0, 1]$, so the entrepreneur keeps $1 - s$.

Baseline utilities from the equity split are:

$$u(s) = p(e, a)(1 - s)X_h - c(e), \quad (1)$$

$$v(s) = p(e, a)sX_h - g(a). \quad (2)$$

The paper calls the decreasing Pareto-efficient segment of the u - v locus the **equity frontier**, written as

$$u = c(v). \quad (3)$$

Because the venture capitalist must pay the investment cost I , deal utilities are

$$U(s) = u(s), \quad (4)$$

$$V(s) = v(s) - I. \quad (5)$$

The associated **contract frontier** is

$$U = C(V) = c(V + I). \quad (6)$$

Interpretation. The equity frontier is about incentive-compatible surplus sharing before subtracting the check that the investor writes. The contract frontier is the actual bargaining set after the investment cost is taken into account.

2.5 Efficient sharing rule

There is a unique interior share

$$s^\# = \arg \max_s [u(s) + v(s)] \quad (7)$$

that maximizes total surplus from the relationship.

At that point the slope of the frontier is -1 , meaning that one extra unit of utility for one side costs exactly one unit for the other side.

Interpretation. If the investor gets *more* than $s^\#$, founders are under-incentivized. If the investor gets *less* than $s^\#$, investors are under-incentivized.

2.6 Search-market objects

The market contains a measure M_e of entrepreneurs and M_v of venture capitalists. The paper defines

$$y \equiv \frac{M_v}{M_e} \quad (8)$$

as the degree of capital-market competition.

A higher y means more investors per entrepreneur, hence a more competitive capital market from the entrepreneur's perspective.

The matching function implies arrival rates

$$q_v(y) = \frac{x(M_e, M_v)}{M_v}, \quad (9)$$

$$q_e(y) = \frac{x(M_e, M_v)}{M_e} = yq_v(y). \quad (10)$$

Interpretation.

- $q_e(y)$ rises with y : entrepreneurs find investors faster when capital is plentiful.
- $q_v(y)$ falls with y : each investor finds deals more slowly when many investors compete for projects.

2.7 Outside options and delay

Search frictions generate outside options U^o and V^o for entrepreneurs and venture capitalists. These are not exogenous bargaining threats; they are endogenous values of going back into the market and searching again.

The relevant discount rate is $r > 0$.

2.8 Entry cost and transparency

Two key market characteristics enter later:

- **Entry cost** k : the fixed cost of entering the venture-capital market.
- **Transparency** χ : a market characteristic that improves matching efficiency. The paper writes the matching technology as $x(M_e, M_v, \chi)$.

Interpretation. A market can become more competitive either because entry is easier or because investors expect higher returns. It can also become more efficient because matching becomes easier.

2.9 Valuation objects

Given investment I and investor share s :

$$L \equiv \frac{I}{s} \quad \text{post-money valuation,} \quad (11)$$

$$G \equiv L - I = \frac{I(1-s)}{s} \quad \text{pre-money valuation.} \quad (12)$$

The paper also distinguishes:

- **total value created** $V^d + U^d$, which nets out effort and investment costs,
- **market value** pX_h , the interim sale value once sunk costs are already paid,
- **success probability** $p(e, a)$.

Interpretation. A high valuation does *not* necessarily mean a high-NPV firm. It may simply mean the entrepreneur has strong bargaining power and the investor accepts a small ownership share.

2.10 Screening extension

In the screening section, projects can be high or low quality. To avoid notation clash with the baseline success probability, it is helpful to denote by π the probability that a project is high quality (the paper reuses p in that section). Screening costs the venture capitalist $C > 0$.

3 Models and empirical specifications (all equations + interpretation)

3.1 Financial contracting and the double-sided incentive problem

The baseline utilities are

$$u(s) = p(e, a)(1 - s)X_h - c(e), \quad (13)$$

$$v(s) = p(e, a)sX_h - g(a). \quad (1)$$

Interpretation.

- Raising s improves the venture capitalist's incentive because she captures more upside.
- But the same increase lowers the entrepreneur's residual claim and weakens founder effort.
- Therefore the first-best “give more to whoever pays the money” logic is wrong in this environment.

3.2 From the equity frontier to the contract frontier

The efficient share satisfies

$$s^\# = \arg \max_s [u(s) + v(s)]. \quad (2)$$

Deal utilities are

$$U(s) = u(s), \quad (14)$$

$$V(s) = v(s) - I. \quad (3)$$

If the equity frontier is $u = c(v)$, then the contract frontier is

$$U = C(V) = c(V + I). \quad (4)$$

Interpretation. The contract frontier is the feasible set of Pareto-efficient contracts once the investor's cash outlay is recognized. Bargaining is simply a choice of one point on this frontier.

3.3 Generalized Nash bargaining

The entrepreneur and investor bargain over a point on the contract frontier. Let U^o and V^o be their outside options. The Nash product is

$$[V - V^o]^\eta [C(V) - U^o]^{1-\eta}, \quad \eta \in (0, 1). \quad (15)$$

Defining $\beta \equiv \eta/(1 - \eta)$, the bargaining first-order condition is

$$\beta = -C'(V^d) \frac{V^d - V^o}{C(V^d) - U^o}, \quad U^d = C(V^d). \quad (5)$$

Interpretation. Bargaining power does not pin down a single split mechanically. Instead, it interacts with the *shape* of the contract frontier. Stronger outside options move the agreed point along that frontier.

3.4 Search market and endogenous outside options

With search frictions, an entrepreneur's outside option is the discounted value of finding another investor:

$$U^o = \frac{q_e(y)}{q_e(y) + r} U^d. \quad (6)$$

Equivalently,

$$rU^o = q_e(y)(U^d - U^o). \quad (7)$$

Similarly for the venture capitalist,

$$rV^o = q_v(y)(V^d - V^o). \quad (8)$$

Interpretation. Outside options are just deal values minus the expected cost of waiting. Faster matching compresses the gap between deal utility and outside-option utility.

3.5 Capital-market equilibrium

Combining bargaining and search gives the paper's key equilibrium condition:

$$\beta \frac{r + q_v(y)}{r + q_e(y)} = -C'(V^d) \frac{V^d}{C(V^d)}. \quad (9)$$

Interpretation.

- The left-hand side is a *search-friction object*: whose waiting cost is more severe.
- The right-hand side is an *incentive-sharing object*: how steep the contract frontier is at the chosen deal.
- Equilibrium is the point where bargaining power implied by market scarcity is consistent with incentive-compatible surplus sharing.

3.6 Baseline comparative statics with respect to capital-market competition

The paper’s first main proposition says that as y rises:

- the venture capitalist’s share s falls,
- the venture capitalist’s deal and outside-option utilities fall,
- the entrepreneur’s deal and outside-option utilities rise,
- total value created $V^d + U^d$ first rises and then falls.

Interpretation. More competition for deals helps founders. But helping founders too much can reduce the investor’s incentive to advise, monitor, and support the firm. The model therefore predicts an interior “best” competition level for value creation.

3.7 Valuation formulas and why valuation is not the same as value

Given investor share s ,

$$L = \frac{I}{s}, \tag{10}$$

$$G = L - I = \frac{I(1 - s)}{s}. \tag{11}$$

Since s falls when y rises, both L and G rise with capital-market competition.

Interpretation. The paper’s critique of industry valuation language is sharp: a high post-money valuation can just mean that investors accept a small share because they face strong competition. It need not mean the project’s true surplus is high.

3.8 Market value and success probability

The interim market value is

$$\text{Market value} = p(e, a)X_h. \tag{12}$$

With a CES success technology,

$$p(e, a) = [\gamma a^\kappa + (1 - \gamma)e^\kappa]^{1/\kappa}. \tag{13}$$

Unless the two efforts are perfect substitutes, the paper shows that market value and success probability are typically also hump-shaped in y .

Interpretation. The strongest market for founders is not necessarily the best market for firm success. Too much investor competition can lower the investor’s stake enough to reduce value-adding effort.

3.9 Endogenous entry of venture capitalists

In the long run, investors enter until their outside option equals the fixed entry cost:

$$V^o = k. \tag{14}$$

This pins down a unique long-run competition level y for each entry cost k .

Interpretation. Free entry makes competition endogenous, but it does *not* guarantee efficient incentive sharing. Entry stops when private profits are driven to the entry-cost benchmark, not when start-up surplus is maximized.

3.10 Changes in investment profitability

An increase in investment profitability is modeled as an outward shift of the frontier through higher X_h .

Short run:

- y is fixed,
- both sides' deal values rise,
- the venture capitalist's equity share s falls.

Long run:

- entry occurs,
- y rises,
- V^d , U^d , and U^o rise,
- V^o is driven back to k ,
- s falls further.

The paper also shows that higher profitability raises total value and valuations in both the short and long run.

Interpretation. This is the boom mechanism: good investment prospects attract more capital, which improves founder terms and speeds financing, but it also eventually weakens investor-side incentives.

3.11 Changes in entry costs

A lower entry cost k raises the equilibrium level of capital-market competition y .

Interpretation. Easier entry works much like an increase in capital supply. It improves founders' terms, but because competition can overshoot the efficient point, it may lower real value creation when the market is already very competitive.

3.12 Changes in transparency

Let the matching technology depend on transparency χ through $x(M_e, M_v, \chi)$.

Short run.

- If $y < 1$ and investors are the scarce side of the market, more transparency strengthens the investor's position in bargaining and increases V^d and s .
- If $y > 1$, the reverse holds.
- In all cases, matching gets faster for both sides.

Long run.

- V^o stays fixed at k by free entry,
- V^d and s fall,
- U^d and U^o rise,
- matching becomes faster for both sides.

Interpretation. In the long run, transparency is unambiguously good for entrepreneurs and for start-up value because it reduces delay and intensifies effective competition.

3.13 Screening extension

Suppose only high-quality projects are worth funding. Let π be the probability that a project is high quality and let screening cost C reveal quality before investment.

Expected utility from screening is

$$\pi V^d + (1 - \pi)V^o - C, \quad (15)$$

while investing without screening gives

$$\pi V^d - (1 - \pi)I. \quad (16)$$

Therefore screening is optimal when

$$C \leq (1 - \pi)(V^o + I). \quad (17)$$

Interpretation. Screening is more attractive when bad projects are common, investment is large, and the investor's outside option is strong. Since V^o is higher when competition is weak, investors screen more in cold markets and less in hot markets.

4 Identification and credibility (what makes the model persuasive)

4.1 Why the paper departs from the “competitive VC market” benchmark

The paper’s credibility starts with a realistic objection to the standard benchmark. In venture capital, it is not reasonable to assume that entrepreneurs always extract all surplus. During hot markets, investors compete aggressively and founders get better terms; during cold markets, the opposite happens. The model builds that institutional fact directly into equilibrium through search and bargaining.

4.2 Why ownership shares matter for real efficiency

The argument is not merely that bargaining changes who gets paid. It changes incentives. Since both founder and investor effort are productive, bargaining power affects the success probability and the total value of the firm. This is why the model has real content instead of being a pure transfer story.

4.3 Why the model is internally disciplined

Several structural restrictions keep the theory tight.

- The contract frontier is strictly concave, so the bargaining solution is unique.
- There is a unique efficient sharing rule $s^\#$.
- The matching technology implies clean monotonicities in outside options through $q_e(y)$ and $q_v(y)$.
- Free entry imposes the zero-profit condition $V^o = k$, which sharply disciplines long-run comparative statics.

Interpretation. The paper is not just telling a verbal story about “hot markets.” It shows exactly how scarcity, bargaining, incentives, and entry fit together in one system.

4.4 Why free entry does not fix the problem

A major persuasive feature is that the paper does *not* claim competition is always bad or always good. Instead, it shows that free entry can leave the market either below or above the value-maximizing competition level because individual entrants ignore the contracting externality they impose on other relationships.

Economic lesson. Private entry equalizes private returns, not social efficiency.

4.5 Robustness to richer contracts

The baseline model sets $X_l = 0$, so the contract is just an equity split. Section 5 then allows a safe payoff $X_l > 0$ and later wage-like transfers. The main message survives:

- more competition still shifts bargaining power,
- the mix of debt, equity, and transfers changes,

- but the central comparative statics for utilities, outside options, and matching remain.

Interpretation. The paper is not hanging entirely on the knife-edge assumption that contracts are pure equity.

4.6 Robustness to broader market structure

The paper also shows that its results continue broadly when:

- entrepreneurs as well as venture capitalists can enter endogenously,
- matching is reconsidered beyond the strict constant-returns benchmark,
- project screening is added before financing.

This matters because it shows the logic is not driven by a single special-case timing assumption.

4.7 Where the paper is strongest empirically

Although the paper is theoretical, it generates clear sign predictions for ownership shares, valuations, matching times, screening intensity, and success rates. It then lines these up with evidence such as:

- Gompers and Lerner's evidence that capital inflows raise valuations,
- anecdotal evidence from the Internet boom and bust,
- Bengtsson et al.'s evidence that screening is weaker in hot markets.

Interpretation. The paper is persuasive because it produces observables with clear directional predictions rather than only abstract welfare claims.

4.8 Main limitations

The assumptions are still strong.

- Agents are risk neutral.
- Contracts are highly stylized relative to real VC contracts with control rights, liquidation rights, and contingencies.
- Beliefs are homogeneous, so bubbles enter only through perceived profitability shifts rather than disagreement.
- Venture capitalists are treated as a representative class rather than layered organizations with limited and general partners.

Interpretation. The model is best viewed as a clean framework for how market structure affects bargaining and incentives, not as a full institutional description of venture-capital contracting.

4.9 Relation to the literature

The paper connects three literatures at once.

- It builds on venture-capital contracting models such as Sahlman, Hellmann, Puri, and Kaplan–Stromberg.
- It imports search-market logic from Diamond–Mortensen–Pissarides style models.
- It contrasts with Michelacci and Suarez by emphasizing not just search externalities but also incentive distortions from ownership shares.

5 Tables and figures

This is a **theory paper**, so there are no regression tables in the usual sense. The main evidence comes from the paper’s figures, propositions, and the sign predictions collected in Section 7.

5.1 Figure 1: equity frontier and contract frontier

Read:

- The dashed curve is the full utility possibility frontier from all possible equity shares.
- The decreasing Pareto-efficient segment is the *equity frontier*.
- Shifting that frontier left by the investment amount I gives the *contract frontier*.

Economic message. This is the paper’s master diagram. Market conditions do not directly choose effort. They move the bargaining point along the contract frontier, which changes the equity split and therefore the incentive balance.

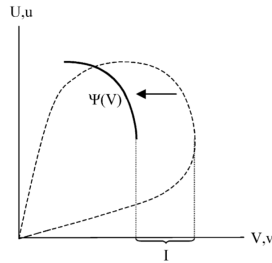


Fig. 1. Equity frontier and contract frontier for the Cobb-Douglas technology. The dashed curve represents the utility possibility frontier (UPF) depicting the entrepreneur’s and venture capitalist’s utilities from equity shares, u and v , for all possible equity shares $s \in [0, 1]$. The decreasing segment of the UPF is the equity frontier. It contains all utilities u and v associated with a Pareto-optimal equity share. Shifting the equity frontier to the left by the investment amount I yields the bold contract frontier $\Psi(V)$, which depicts the entrepreneur’s and venture capitalist’s utilities from Pareto-optimal contracts, $U = \Psi(V)$ and V , respectively.

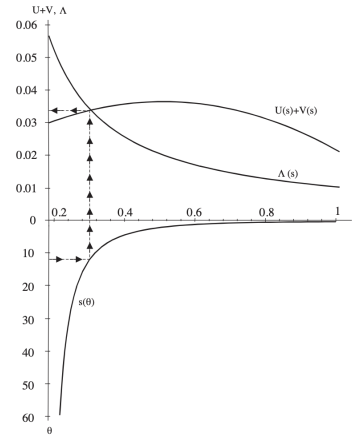


Fig. 2. Total value created in the start-up, $U + V$, and post-money valuation A as a function of the venture capitalist’s equity share s , which in turn is a function of the level of capital market competition θ . The functional and numerical specifications are given in Section 3.2. The set of Pareto-optimal equity shares is $s \in [0.175, 1]$. $U + V$ and A intersect at $s \approx 0.29$, or $\theta \approx 12.4$. If $\theta < 12.4$ the valuation A understates the true value created in the start-up, while if $\theta > 12.4$ it overstates it. The graph of $s(\theta)$ has been curtailed at $\theta = 60$ for expositional purposes.

5.2 Proposition 1 and Figure 2: value creation versus valuation

Read:

- As capital-market competition y rises, the investor’s share s falls and valuations rise.

- But total value created, $U^d + V^d$, is inverted-U in y .
- In the paper's numerical example, the post-money valuation intersects true value around $s \approx 0.29$, corresponding to $y \approx 12.4$.

Economic message. The eye-catching point is that valuation and value are different objects. In low-competition markets valuation can *understate* true value, while in very competitive markets it can *overstate* true value.

5.3 Proposition 2: pre- and post-money valuations

Read:

- Both pre-money and post-money valuation rise monotonically with competition.
- The formulas $L = I/s$ and $G = I(1 - s)/s$ make this mechanical once s falls with y .

Economic message. Higher valuations in hot markets do not automatically mean better projects. They may simply reflect a thin bargaining position for investors.

5.4 Proposition 3: success probability and market value

Read:

- If founder and investor efforts are not perfect substitutes, success probability and interim market value are also hump-shaped in competition.
- Too little competition gives founders too little stake; too much gives investors too little stake.

Economic message. The model predicts that real outcomes can deteriorate at both extremes of market tightness.

5.5 Figure 3 and Propositions 5–7: boom-bust dynamics

Read:

- A profitability shock shifts the frontier outward.
- In the short run, with the number of investors fixed, both sides' deal utilities rise and investor ownership falls.
- In the long run, entry raises competition further, valuations rise, founders find financing faster, and investors take longer to find deals.

Economic message. Figure 3 is the paper's stylized picture of the Internet boom and bust. It shows how a profitability shock first changes the value of deals and then, through entry, changes market structure itself.

5.6 Figures 4 and 5: safe payoff and wage extensions

Read:

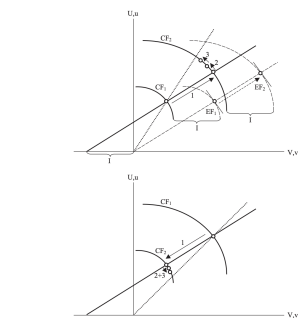


Fig. 3. Short- and long-run effects of an increase (upper picture) and decrease (lower picture) in investment profitability. The upper picture depicts both the change in the (dashed) equity frontier and (solid) contract frontier, while the lower picture depicts only the change in the contract frontier. An increase in investment profitability shifts both the equity and contract frontier outward, from EF_1 to EF_2 and from CF_1 to CF_2 , respectively. Effects 1 and 2 together constitute the short-run effect. Effect 3 depicts the additional adjustment in the venture capitalist's equity share that occurs in the long run. Effects 1-3 together constitute the total, i.e., long-run, effect.

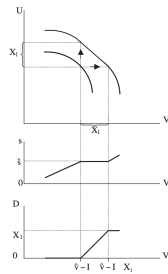


Fig. 4. Contract frontier and optimal contract if there is a safe payoff X_l . Adding a safe payoff shifts the contract frontier outward. The contract frontier with $X_l > 0$ is constructed from that with $X_l = 0$ as follows: The segment where $s < l$ is shifted upward, and the segment where $s > l$ is shifted rightward by X_l . The shifted segments are connected by a straight line with slope minus one. In the picture, s and l denote the venture capitalist's and second-best equity share, respectively. U and V are the entrepreneur's and venture capitalist's utilities from Pareto-optimal contracts, respectively, and $l - f$ is the venture capitalist's utility under the contract ($l, D = 0$), where $D = 0$ is the amount of debt held by the venture capitalist. The middle and lower pictures show how s and D vary along the (new) contract frontier.

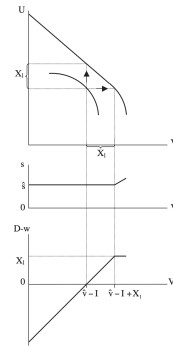


Fig. 5. Contract frontier and optimal contract if there is a safe payoff X_l and wage payments w are potentially unlimited. The construction of the contract frontier with $X_l > 0$ and $w > 0$ is similar to Fig. 4, except that the left segment where $s < l$ —i.e., where the venture capitalist's equity share is below the second-best equity share—is replaced by a straight line with slope minus one. In the picture, U and V are the entrepreneur's and venture capitalist's utilities from Pareto-optimal contracts, respectively, and $l - f$ is the venture capitalist's utility under the contract ($l, D = w = 0$), where $D = 0$ is the amount of debt held by the venture capitalist. (Wages and debt are perfect substitutes, which implies that only the difference $D - w$ is uniquely determined.)

- Adding a safe payoff $X_l > 0$ introduces debt-like transfers.
- Allowing wages adds another non-distortionary transfer device.
- The frontier develops a linear region where utility can be shifted without changing the equity split.

Economic message. Richer contracts can soften the distortion from bargaining, but they do not eliminate the underlying importance of market structure.

5.7 Section 7 empirical implications: what should move in the data

Read:

- VC ownership shares should fall when capital supply, profitability, or transparency rise, and increase when entry costs rise.
- Valuations should rise with capital supply, profitability, and transparency, and fall with entry costs.
- Entrepreneur search time should fall when competition, profitability, or transparency rise.
- VC screening should be stronger when competition is weak.

Economic message. The model's real empirical value is the sign structure. It tells us which contractual and operational outcomes should co-move with hot and cold venture-capital markets.

6 Short summary

Inderst and Muller build a model in which venture-capital market conditions affect start-ups through bargaining and incentives. More available capital improves entrepreneurs' outside options, lowers investor ownership shares, and raises valuations. But because both founders and investors must exert effort, value creation is not monotone in investor competition: it is highest at an interior ownership split that balances the two incentive problems. In the long run, shocks to profitability, entry costs, and transparency change market structure through entry and matching. The paper's

central insight is that capital supply affects not only pricing but also the real efficiency of start-up finance.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that venture-capital market structure is a first-order determinant of ownership, valuation, and value creation. A thicker capital market strengthens entrepreneurs' bargaining positions and improves financing terms, but it can also undermine investor incentives if competition becomes too strong. Free entry therefore need not produce efficient start-up finance.

7.2 Economic intuition

Start-up contracts allocate upside. Upside allocation determines effort incentives. Search frictions determine outside options, and outside options determine how upside gets allocated in bargaining. Once these pieces are connected, a change in capital supply or transparency does far more than move prices – it changes the internal governance of the firm and therefore its chances of success.

7.3 Takeaway

This paper is one of the clearest theoretical statements of why “too much money chasing too few deals” is not just a valuation story. It is an incentive story. Hot markets can improve founder terms and speed financing, but they can also push ownership too far away from the value-maximizing split. That is why market competition in venture capital has both distributional and real consequences.